

Using sentiment analysis assisted RAG models for product review summarization, feature evaluation, and user question answering

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ABSTRACT

As the volume of online reviews continues to grow, finding relevant insights has become increasingly challenging and tedious. This study explores the use of a large language model (LLM)-assisted tool to improve the review-browsing experience compared to manual methods. Four key metrics—convenience, quickness, detail, and relevance—were evaluated. Results indicate that the LLM-assisted tool is significantly improved in terms of convenience and quickness compared to manual reading of reviews, while showing comparable performance in the detail and relevance of the information gathered. These findings demonstrate the potential of LLMs to streamline review browsing and provide a foundation for further improvement in delivering high-quality insights.

CCS CONCEPTS

• Information systems → Information retrieval; • Human-centered computing → Empirical studies in HCI; • Artificial intelligence → Natural language processing; Interactive systems.

KEYWORDS

AI-Assisted Tools, Large Language Models, Review Summarization, Human-AI Interaction, Online Reviews, Natural Language Processing, Information Retrieval

ACM Reference Format:

Arjun Devireddy, Kushal Choudhary, Anshuman Das, and Ruoyu Xu. 2018. Using sentiment analysis assisted RAG models for product review summarization, feature evaluation, and user question answering. In *Woodstock '18: ACM Symposium on Neural Gaze Detection, June 03–05, 2018, Woodstock, NY*. ACM, New York, NY, USA, 8 pages. <https://doi.org/XXXXXXX.XXXXXXX>

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Conference acronym 'XX, June 03–05, 2018, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/18/06...\$15.00

<https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

Online product reviews are valuable resources in evaluating the strengths and weaknesses of a product, and can also give insight into customers' preferences. Information gathered from online reviews can often paint a good picture of how future products can be developed, or how future iterations of a product can be improved according to customers' preferences. Potential customers often rely on customer reviews to assess the quality of a product and decide whether to buy a product or not, or look for alternative products. Product designers will similarly examine reviews to find complaints and decide the best ways to improve the product in future iterations. It's evident that online product reviews are an important part of both the product sales and product design processes. But, perusing through reviews manually can be time consuming and have a high cognitive load as the reader has to mentally sort through and develop a consensus of the reviews, so automating this extraction and summarization of insights could be key to more efficient usage of reviews. In our research we propose a new method that utilizes the increasingly popular Large-language model (LLM) in a retrieval-augmented generation process to allow the LLM to generate a digestible summary of product reviews, evaluate a product's features, and answer further user-specified questions about the product and reviews. This should allow users to quickly get a general idea of a product's strengths, weaknesses, and how the product can be improved. Additionally, users can get answers to specific concerns about a product without having to dig through reviews to look for relevant information. To test the usefulness of our model, We perform a user study and examine the convenience and quickness of using our model compared to manually reviews reviews, as well as the level of detail and relevance of the responses produced by the LLM as compared to the information found by manually reading reviews.

2 RELATED WORK

2.1 Sentiment Analysis

Sentiment analysis, a crucial component of natural language processing (NLP), has gained significant attention for its ability to extract and interpret subjective information from text, enabling

businesses to gauge customer opinions and emotions with increasing accuracy. Previous techniques like traditional sentiment analysis models, which often rely on bag-of-words or basic polarity-based methods, tend to lose the context in which a product feature is being critiqued as they filter out words unrelated to sentiments [5, 9]. More modern techniques, such as deep neural networks (DNNs), have significantly advanced sentiment analysis by maintaining the context of entire sentences. Unlike traditional approaches that treat words independently, DNNs—especially architectures like recurrent neural networks (RNNs) and transformers—can capture relationships between words and phrases across a sentence [11]. This allows the model to consider the nuanced interaction between different parts of the text, preserving the context in which specific product features are mentioned [12].

However, these supervised learning methods face limitations, including the scarcity of large, domain-independent labeled datasets and the challenge of interpreting ambiguous and implicit sentiment expressions [1]. Even more recent AI methods, such as Large Language Models (LLMs) like BERT and GPT, have addressed these challenges by leveraging the semantic context embedded within pre-trained language models. These models can be fine-tuned with smaller labeled datasets using transfer learning, enabling them to better capture subtle emotional expressions and achieve state-of-the-art performance [6]. To fully capture the nuanced opinions in product reviews, it is essential to combine advanced sentiment analysis techniques with feature extraction [16], enabling models to not only classify the overall sentiment but also associate specific sentiments with individual product features for a more comprehensive understanding of customer feedback.

2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a crucial technique that enhances the performance of Large Language Models (LLMs) by integrating external knowledge to generate more accurate and high-quality responses. One significant advancement in this field is the combination of RAG with knowledge graphs (KGs) for customer service question-answering tasks. By using KGs to preserve both intra-issue structures and inter-issue relationships, this method substantially improves retrieval accuracy and response generation quality. Deployed in LinkedIn’s customer support, this approach has reduced median issue resolution time by 28.6 percentage, demonstrating how domain-specific knowledge integration can streamline customer service processes and enhance response efficiency [18].

In addition to specific applications in customer service, RAG has also shown promise in addressing challenges related to automatic text summarization (ATS). Traditional methods for summarization, particularly extractive and abstractive approaches, struggle with the ever-increasing volume of textual data. A comprehensive survey on ATS highlights these limitations, emphasizing that current models are far from human-level performance. RAG has the potential to bridge this gap by leveraging external knowledge to improve the abstraction quality of summaries, making it a valuable tool for handling vast text corpora [3]. Furthermore, transparency in RAG models has been addressed through frameworks like RAG-Ex, which explains why certain responses are generated, thereby

enhancing user trust and aligning model outputs with human expectations [15].

Finally, a recent survey on the intersection of RAG and LLMs discusses how RAG helps mitigate issues like hallucinations and outdated information in LLMs by incorporating up-to-date, external knowledge. This combination of retrieval and generation significantly improves the relevance and reliability of the generated content, particularly in AI-generated content (AIGC) applications. By augmenting LLMs with current knowledge sources, RAG ensures that the models provide high-quality responses across a variety of tasks [4].

2.3 Gaining insights from product reviews

The process of generating summaries and tips from online reviews has been studied previously [7, 8], using methods like using sentiment analysis with deep learning [2]. However, deep learning based extraction is typically not responsive to user queries the way LLMs are and instead generate a static list of product features or assessments from a set of reviews. In the state-of-the-art, Ensemble Deep Learning Models (EDLM) [14] have been used to extract and classify specific product features from text-based reviews. Techniques such as conjoint analysis models and Kano models can be used to then develop appropriate product improvement strategies, and can also analyze and prioritize those extracted product features [13, 14]. Additionally, most review platforms offer users the ability to give both numerical (typically Star ratings) and textual reviews. There may be inconsistencies between these numerical and textual reviews, so to take this into account, an Inconsistent Ordered Choice Model can be used, which may more accurately measure customer’s preferences about certain product features. Based on these customer preferences, sentiment-based models such as sentiment-based importance-performance analysis (SIPA) can analyze and categorize product features to guide product development [17]. The introduction of sentiment analysis into the model likely leads to product improvement guidance that is more adaptive to customer preferences compared to previous IPA models [17], though this still needs to be further studied quantitatively. Research has also been done into automatically filtering reviews before analyzing them. By discarding bad reviews, a smaller higher-quality pool of reviews should be left, and these should yield a better final result after running any model [13, 19]. An alternative approach to the product feature-based analysis mentioned so far is to observe customers and try to gain an understanding of their preferences, wants, and expectations[20]. This can be done by correlating online reviews to customer journey maps (CJM). The CJM mapping identifies a behavior-based Voice of Customer (VOC), and this VOC has been argued to offer broader insights on product development than feature-centric methods [10]. Research on these techniques of filtering, product feature extraction, categorization and prioritization, and sentiment-based analysis is at the core of gaining insights from online product reviews to guide product improvement and development.

However, the usage of Large-language models for the purpose of summarizing and finding specific information from online reviews has not yet been studied, and thus gives our research an unique perspective.

3 METHOD

Since LLMs have demonstrated potential in tasks such as text summarization and question answering [3]. Our research attempts to build on this by exploring the effectiveness of large language models (LLMs) in providing meaningful insights to users about products by leveraging Amazon reviews.

Oftentimes, product pages on online stores feature a large number of reviews. Each one comes from a different perspective, with different impressions of the product, and each one covers a different set of features related to the product. For example, on a product page for a laptop, some reviews might complain about the battery life, some will praise the display quality, and so on. Different reviews will have differing opinions on all the different features.

Potential shoppers often have a hard time going through these reviews to find answers to questions they may have about the product or get a well rounded understanding of benefits and shortcomings. Product designers also often refer to product reviews in a similar way to see how they can improve the product and address common issues.

To aid these shoppers and designers, we aim to test how effective LLMs coupled with retrieval-augmented generation are at summarizing product reviews to generate a more digestible and systematic overview of the product. We also hope to see how chatting with the LLM can answer user questions about the product more quickly and directly than having to manually search through reviews.

Specifically, we address the following research questions:

Can LLMs using retrieval-augmented generation (RAG) help users better understand product reviews? We investigate whether the combination of LLMs and RAG can effectively retrieve and summarize relevant product information from large datasets of Amazon reviews to enhance user comprehension. Our hypothesis is that LLMs using retrieval-augmented generation (RAG) will provide a more efficient method of understanding product reviews for users compared to manual methods, as measured by speed of information gathering, specificity of answers, usability and convenience. We expect users to obtain information in a shorter time, find more specific answers, and find our system to be more usable and convenient than manual methods.

Does adding sentiment-based tags to reviews improve the quality of LLM-generated insights? Since past studies have shown that sentiment-labeled datasets better help capture emotions, We evaluate whether performing sentiment analysis and tagging reviews before using them in the RAG model enhances the accuracy, contextual relevance, and overall quality of the generated responses [6].

Additionally, we expect that adding sentiment-based tags to the reviews will improve the quality of the insights generated by the LLM. In our approach, we integrated an LLM with a user interface that retrieves reviews to answer user queries. Sentiment-based tags were applied to the reviews in a pre-processing step to assess their potential impact on improving the insights generated by the model.

3.1 Document and Vector database creation

The Langchain framework’s integration with the Chroma vector database allowed the creation of vector databases for each product’s reviews. Initially, separate databases were created for both

general feedback summaries and sentiment-tagged feedback, but for the final prototype, the sentiment-tagged database was used. Each product’s vectorized review data was indexed, enabling fast retrieval based on a similarity search. A general product summary retriever was constructed to retrieve relevant reviews based on keyword-based or context-specific queries and a separate sentiment-specific retriever was designed to retrieve sentiment-tagged reviews, allowing us to highlight aspects of customer feedback that were explicitly positive or negative.

3.1.1 Sentiment-tagged Feedback Vector Database. For each product, we established a feedback vector database. Initially, customer reviews were pre-processed to ensure they were clean and relevant, which involved removing extraneous characters and segmenting longer reviews into smaller, coherent chunks. Using Google Generative AI Embeddings (text-embedding-004 model), we transformed these reviews into dense vector representations. When users submitted queries, the system could efficiently retrieve the most relevant reviews by comparing the vector embeddings of the input query with those in the database to find the reviews most relevant to the query and return them as context to the LLM, thus providing more relevant summaries and answers based on customer feedback.

3.1.2 Sentiment-tagged Vector Database. In addition to the general feedback database, we created a sentiment-specific vector database for each product. As sentiment tagging makes a clearer distinction of whether a review is praise or complaint it can assist in the retrieval and LLM generation process. Reviews in this database were first tagged with sentiment classifications (e.g., very positive, positive, neutral, negative, very negative) using a sentiment-analysis model prior to vectorization. This tagging enabled us to conduct more nuanced searches that took sentiment into account, allowing users to highlight specific aspects of customer feedback [6]. For instance, if a user wanted to see a summary of negative reviews about a product’s battery life, the sentiment-specific retriever would prioritize reviews that contained relevant sentiment information leaning negative, and provide more accurate insights into customer satisfaction and concerns.

3.1.3 Retrievers for Effective Feedback Summarization. To facilitate effective feedback summarization, we developed two distinct retrievers. General Product Summary Retriever: This retriever accessed the general feedback database to pull relevant reviews based on keyword or context-based queries. It ensured that users received a comprehensive overview of customer experiences related to specific product attributes. Sentiment-Specific Retriever: This retriever utilized the sentiment-specific database to extract reviews that matched the input query and sentiment. By filtering reviews based on both content relevance and sentiment polarity, it allowed for targeted feedback analysis, helping users understand what features were praised or criticized. These retrievers worked synergistically within the broader retrieval-augmented generation (RAG) framework to pull relevant customer feedback from the vector databases. This structured approach ensured that the LLM could produce nuanced, context-aware feedback summaries for each product, thereby enhancing the overall quality of the insights generated.

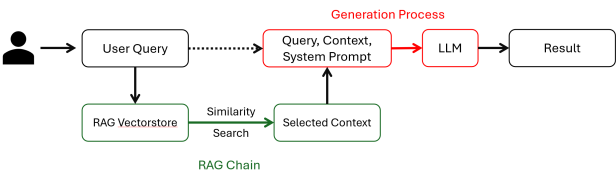


Figure 1: Example of RAG process

3.2 LLM integration using Langchain

To streamline the interaction between user prompts, product descriptions, and review data, we leveraged the powerful chain-building capabilities of Langchain. This framework allowed us to create a seamless integration of various components, facilitating an efficient flow of information between the user, the LLM, and the underlying data sources. The system prompt was meticulously crafted to guide the large language model (LLM) in generating high-quality product summaries that are both informative and contextually relevant. This process involved synthesizing information from retrieved reviews while incorporating sentiment analysis to provide a nuanced understanding of customer feedback. The LLM utilized in our study is the Google Generative AI, specifically the Gemini-1.5 Flash model. This advanced model is capable of generating detailed and coherent summaries by processing not only the product descriptions but also the most pertinent customer reviews. The system prompt included explicit instructions designed to elicit key insights from the LLM. These insights focused on highlighting both positive and negative sentiments expressed by customers, enabling a balanced perspective on product performance. The retrieval process was fundamentally supported by Chroma’s vector database, which efficiently indexed and stored the review data. This database facilitated rapid access to relevant reviews based on user queries, ensuring that the LLM was provided with the most applicable context for each summarization task. By employing similarity search techniques, we were able to retrieve reviews that aligned closely with the user’s prompt, enhancing the relevance of the information presented. Once the relevant reviews were retrieved, the LLM generated a response that was further enriched by these insights. The integration of retrieved data from the vector store into the LLM-generated output ensured that the final summaries were not only comprehensive but also grounded in actual customer experiences. This dual approach—combining the analytical capabilities of the LLM with the rich data sourced from customer reviews—resulted in summaries that offered valuable insights into product strengths and weaknesses.

3.3 Summary Generation and Chatbot

The LLM consisted of two features. The first feature allows users to ask specific questions related to the product, enabling them to gain targeted information based on their interests or concerns. The second feature of the LLM focuses on summarizing the reviews by identifying key strengths, weaknesses, and areas for improvement. By integrating these two features, the LLM not only facilitates

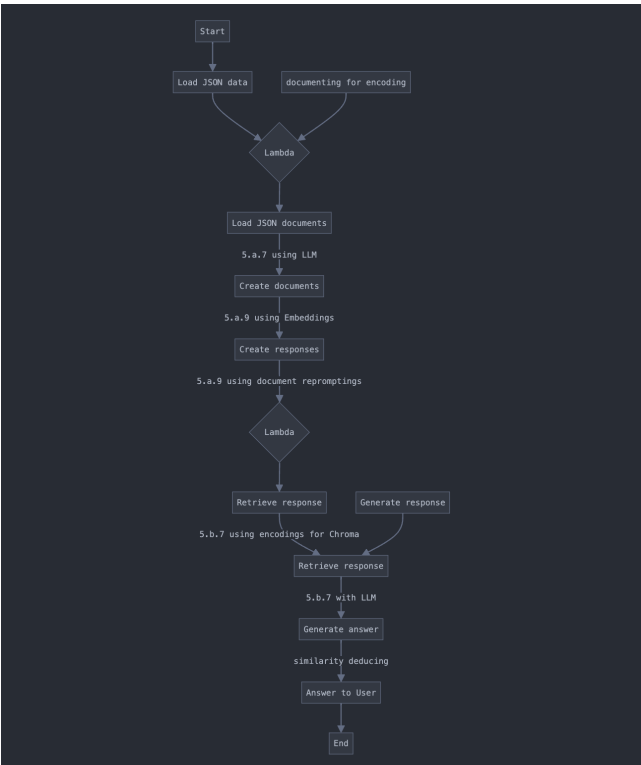


Figure 2: System Flowchart: From Document Creation to LLM-Generated Response

a seamless user experience but also empowers consumers with well-rounded insights.

3.4 User Interface using Streamlit

To facilitate user interaction with the large language model (LLM) and streamline the process of retrieving product insights, we developed a simple user interface (UI) using Streamlit. This UI is designed to mimic a chatbot format, providing an intuitive platform for users to pose questions about products.

The Streamlit application features a side panel where users can select the product they wish to inquire about, ensuring that the responses are relevant to their chosen items. Once a product is selected, users can type their inquiries into a text input field. The UI then presents the LLM-generated responses in a clear display area. This design allows for a streamlined interaction, making it easy for users to navigate the interface without encountering technical barriers. Upon receiving a user query, the UI processes the input, sending it to the backend where the LLM retrieves relevant information from the pre-processed Amazon reviews. The response is then generated and returned to the UI for display. Figure 2 and 3 illustrate an example of the UI, showcasing a sample question and response interaction.

4 USER STUDY

We conducted a user study to test the effectiveness of our LLM for product research. Participants completed research tasks for

InsightGen: AI-Powered Feedback Summarizer

Enter prompt:

Enter

General Product Summary

Audio-Technica ATH-M30 Headphones: Customer Review Summary

Strengths:

- Exceptional Sound Quality:** Customers consistently praise the sound quality, describing it as "crisp," "clean," "balanced," and "unbelievable." Many note that they can hear details in their music they haven't noticed before, even with "untrained ears."
- Value for Money:** The headphones are consistently lauded as a fantastic value, outperforming much more expensive options. Customers are impressed with the quality they get for the price point.
- Noise Isolation:** The closed-back design effectively blocks out ambient noise, making them ideal for work, travel, or concentrating in noisy environments. Customers appreciate the ability to focus on their music or work without distractions.
- Comfort:** The headphones are deemed comfortable for extended use, even for multiple hours. The adjustable headband and soft earcups are praised for their comfort.
- Durability:** Many reviewers highlight the durability of the headphones, with some reporting that they have lasted for several years without issue.

Weaknesses:

- Long Cord:** The long cord, while useful in some situations, is seen as a drawback for portability and can be cumbersome for everyday use.
- Lack of Isolation for Phone Calls:** Some reviewers mention that the headphones don't completely block out outside sounds, making them less ideal for phone calls or situations where you need to hear your surroundings.
- Limited Bass:** While the bass is described as "balanced," some reviewers note that it isn't as "booming" as they might prefer.

Areas for Improvement:

- Cord Length:** Offering a shorter cord option could enhance portability and make the headphones more versatile.
- Improved Noise Isolation:** While the headphones are good at noise isolation, some customers expressed a desire for even better noise cancellation.
- Bass Boost Option:** A feature that allows users to adjust the bass level would cater to those who prefer a more powerful bass response.

Specific Features Praised:

- Gold-plated connector:** Reviewers appreciate the quality and durability of the connector.
- Adjustable headband:** The comfort and flexibility of the headband are a big plus.
- Screw-on adapter:** The inclusion of the adapter for 1/4" (6.3 mm) connections makes the headphones compatible with a wider range of devices.

Specific Features Criticized:

- Cord length:** The long cord is repeatedly mentioned as a potential inconvenience.

Overall:

The Audio-Technica ATH-M30 headphones are a highly recommended product for those seeking excellent sound quality at an affordable price. Customers consistently praise the sound quality, comfort, and value for money. While the long cord and limited bass response are potential drawbacks, the headphones excel in noise isolation and overall performance, making them a popular choice for work, travel, and music enjoyment.

Figure 3: Example of Generating a General Product Summary

InsightGen: AI-Powered Feedback Summarizer

Enter prompt:

What do people think about the sound quality?

Enter

General Product Summary

The sound quality of the Audio-Technica ATH-M30 headphones is generally considered to be good for the price, but not exceptional. Many reviewers find the sound to be clear and detailed, with a good balance of highs and lows. However, some find the bass response to be lacking, and others describe the sound as "muddy" or "flat."

Here's a breakdown of the different perspectives:

Positive:

- Good for the price:** Many reviewers emphasize that the sound quality is good considering the price point.
- Clear and detailed:** Several reviewers praise the clarity and detail of the sound, particularly in the mids and highs.
- Good balance:** Many reviewers find the balance of highs, mids, and lows to be pleasing.
- Crisp and punchy bass:** Some reviewers highlight the crisp and punchy bass response, although others find it lacking.

Negative:

- Lacking bass:** A common complaint is that the bass response is not as strong as some would like.
- Muddy or flat sound:** Some reviewers find the sound to be muddy or flat, particularly at higher volumes.
- Not as good as more expensive headphones:** Several reviewers compare the sound quality to more expensive headphones and find it to be inferior.

Overall:

The majority of reviewers agree that the Audio-Technica ATH-M30 headphones offer decent sound quality for the price. However, those seeking exceptional sound quality or a powerful bass response may be disappointed. It's important to consider your individual preferences and budget when making a decision.

Figure 4: Example of User Submitting a Query

four different products, using two distinct methods: manual review browsing and LLM interaction. This dual-method approach allowed us to compare traditional review browsing directly with using retrieval augmented large-language models (LLM), for the purpose of providing valuable insights into the relative efficiency, ease of use, and effectiveness of each method for gathering product information. This study was designed as a within-participant study to allow each participant to compare both methods and to keep likert-style evaluations more consistent between both methods.

4.1 Participants

We recruited 14 participants (ages 18 to 26, mean age = 21.6), consisting of 8 males and 6 females. Participants were recruited using convenience sampling and were screened to ensure prior experience with LLMs/chatbots to facilitate ease of use and engagement with the study's tasks.

4.2 Study design

4.2.1 Training. To ensure participants were comfortable using the LLM and manual review processes, we provided a brief training session prior to beginning the main portion of the study. This training consisted of a demo video and practice questions designed to introduce participants to the LLM’s interface and response capabilities.

Upon completion of the training session, participants were encouraged to ask any questions they had about using the LLM before proceeding to the main study. This training phase ensured that participants were adequately prepared and confident in using the LLM, reducing the likelihood of confusion and improving the reliability of study results.

4.2.2 Procedure and Design. For each product, all participants first engaged in manual review browsing by navigating to the product’s Amazon page and reading through customer reviews to answer specific questions about common product issues, improvement suggestions, comparisons with similar products, and strengths. This step allowed participants to experience a traditional approach to gathering product insights directly from customer reviews and served as a control group.

In the following LLM section, all participants interacted with the specialized LLM that generated product insights based on aggregated customer reviews. They asked the LLM the same product-related questions as in the manual review section. By interacting with the LLM, participants explored how this AI-assisted approach compared to manual browsing in terms of ease and efficiency in obtaining relevant product information.

After completing the tasks for each method for all four products, participants rated their experience using a series of Likert-scale questions. These questions assessed the perceived convenience, speed, detail, and relevance of the information obtained through the manual review browsing method and the interactive LLM method. This structured procedure enabled a thorough comparison of both methods, providing insight into the LLM’s potential as an alternative to traditional review browsing for product research.

4.3 Results

The Wilcoxon signed-rank test was performed to compare the perceived convenience, quickness, detail, and relevance of information obtained through manual review browsing and the LLM-assisted approach across four products.

4.3.1 Convenience. The comparison of convenience between the manual review ($M = 2.50$, $SD = 1.45$) and LLM ($M = 6.36$, $SD = 0.75$) methods revealed a statistically significant difference. $p = 0.001$ suggests that participants perceived the LLM method to be more convenient than the manual review method. The null hypothesis, which posits no difference between the two methods, was rejected, indicating that the LLM was considered more convenient for gathering product insights.

4.3.2 Quickness. A significant difference was found in the perceived quickness of the manual review ($M = 2.64$, $SD = 1.55$) and LLM ($M = 6.57$, $SD = 0.514$) methods. $p = 0.001$ indicates that participants found the LLM approach faster compared to the manual review browsing. This result further supports the notion that the AI-assisted method is more efficient for users in terms of time.

4.3.3 Detail. The Wilcoxon signed-rank test for detail did not show statistical significance between the manual ($M = 4.21$, $SD = 1.89$) and LLM ($M = 5.57$, $SD = 0.76$) methods, with $p = 0.07$ that exceeds the 0.05 significance threshold. This suggests that, despite differences in convenience and speed, we cannot conclude that participants found the level of detail provided by the LLM to be better than going through reviews manually. However, looking at the means of the two groups obtained in this study (4.21 against 5.57), there is still potential that using the LLM can produce more detailed results. Perhaps with a better LLM architecture or a more powerful study with more participants, statistical significance can be shown.

4.3.4 Relevance. No significant difference was observed in the relevance of the information obtained through the manual review ($M = 4.36$, $SD = 1.22$) and LLM methods ($M = 5.35$, $SD = 1.22$). $p = 0.14$ suggests that both methods were equally effective in providing relevant insights about the products. However, as with detail, the mean of the data using the LLM was higher, which shows that within this study, participants did on average feel that the LLM produced more relevant results. As with detail, the LLM is potentially better, but it is inconclusive based on the data in this study.

4.3.5 Summary. In conclusion, the results support the hypothesis that the LLM-assisted approach was perceived as more convenient and quicker than the traditional manual review method. It remains inconclusive whether using an LLM improves the level of detail or relevance of information obtained from reviews, as statistical significance for these metrics has not yet been demonstrated. Further testing, with a better LLM model, or a more robust study design, may present more conclusive findings. These findings highlight the potential of the LLM method as an alternative to manual review browsing, offering improvements in convenience and efficiency without compromising the quality of insights.

Table 1: Wilcoxon Signed-Rank Test Results

	LLM Mean (SD)	Manual Mean (SD)	p-value
Convenience	6.357 (0.745)	2.500 (1.454)	0.001**
Quickness	6.571 (0.514)	2.643 (1.550)	0.001**
Detail	5.571 (0.756)	4.214 (1.888)	0.071
Relevance	5.357 (1.216)	4.357 (1.216)	0.136

5 DISCUSSION

This study evaluated the effectiveness of a large language model (LLM)-assisted tool in enhancing the review-browsing experience compared to manual methods. The results demonstrated that the LLM significantly improved convenience and quickness, offering a streamlined approach for extracting insights from reviews. These findings highlight the potential of LLMs in reducing the cognitive load and time commitment required for users to navigate large datasets, making them a valuable tool in consumer-focused applications.

5.1 Limitations

This study highlighted the potential of LLMs in improving the review-browsing experience; however, several limitations need to be addressed. First, the responses generated by the LLM were not

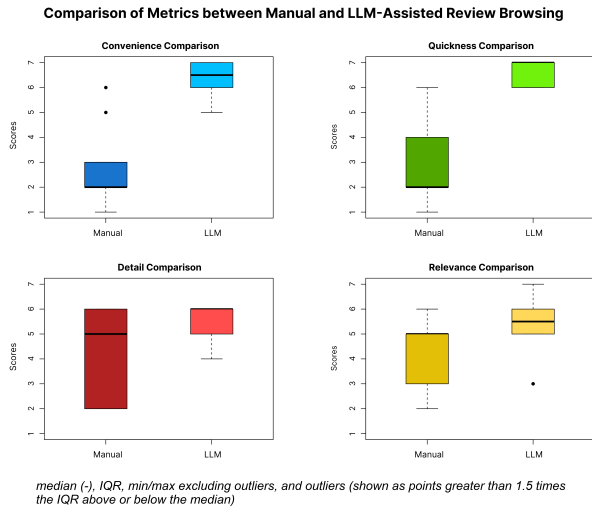


Figure 5: Comparison of Metrics between Manual and LLM-Assisted Review Browsing

always consistent, occasionally leading to varying levels of relevance and detail for similar queries. This inconsistency may have influenced the participants' evaluations, particularly in metrics like detail and relevance. Additionally, to maintain participant engagement and avoid overwhelming them, the study was designed with a limited set of questions. While this ensured a smoother experience, it may have constrained the depth and diversity of insights gathered. Addressing these limitations will be crucial to improving the tool's robustness and user experience.

5.2 Future Work

Future iterations of this study will focus on several enhancements to address the identified limitations. First, experimenting with different prompt structures and phrasing will be prioritized to improve the consistency and quality of the LLM's responses. Second, the system will be expanded to handle more complex queries, such as those comparing multiple products, providing users with richer and more actionable insights. Lastly, conducting qualitative analysis will offer deeper insights into how participants perceive and interact with the tool, providing a more nuanced understanding of its strengths and areas for improvement. These advancements aim to further refine the LLM-assisted review browsing tool and maximize its utility for users.

6 CONCLUSION

This study explored the potential of a large language model (LLM)-assisted tool to improve the online review-browsing experience, comparing its performance to manual methods across four key metrics: convenience, quickness, detail, and relevance. The findings demonstrated that the LLM significantly enhanced convenience and quickness, making it a valuable tool for streamlining the review analysis process. However, the tool showed mixed results in delivering consistent, detailed, and relevant responses, highlighting

areas for improvement. While the results underscore the promise of LLMs in reducing the time and effort required to extract insights, they also reveal the challenges of ensuring consistency and robustness in AI-generated outputs. These limitations, combined with the constrained scope of this study, suggest that further refinement and experimentation are necessary to fully harness the capabilities of LLMs in this context. In conclusion, the LLM-assisted tool represents a significant step forward in utilizing AI to enhance user experiences in data-dense environments. By addressing its limitations and expanding its functionalities, future iterations of the tool can provide even greater value, offering more reliable and comprehensive insights for users navigating complex datasets such as online reviews.

ACKNOWLEDGMENTS

[FILL FOR P6]

To Robert, for the bagels and explaining CMYK and color spaces.

REFERENCES

- [1] Jyotsna Anthal, Bhavna Sharma, and Jatinder Manhas. 2023. Optimization-enabled deep learning for sentiment rating prediction using review data. *Service Oriented Computing and Applications* 17, 1 (01 Mar 2023), 39–58. <https://doi.org/10.1007/s11761-023-00357-9>
- [2] Jihua Cao, Jie Li, Miao Yin, and Yunfeng Wang. 2023. Online Reviews Sentiment Analysis and Product Feature Improvement with Deep Learning. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 22, 8, Article 203 (aug 2023), 17 pages. <https://doi.org/10.1145/3522575>
- [3] Wafaa S. El-Kassas, Cherif R. Salama, Ahmed A. Rafea, and Hoda K. Mohamed. 2021. Automatic text summarization: A comprehensive survey. *Expert Systems with Applications* 165 (2021), 113679. <https://doi.org/10.1016/j.eswa.2020.113679>
- [4] Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. 2024. A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. Association for Computing Machinery, New York, NY, USA, 6491–6501. <https://doi.org/10.1145/3637528.3671470>
- [5] Xing Fang and Justin Zhan. 2015. Sentiment analysis using product review data. *Journal of Big Data* 2, 1 (16 Jun 2015), 5. <https://doi.org/10.1186/s40537-015-0015-2>
- [6] Gihan Gamage, Daswin De Silva, Nishan Mills, Daminda Alahakoon, and Milos Manic. 2024. Emotion AWARE: an artificial intelligence framework for adaptable, robust, explainable, and multi-granular emotion analysis. *Journal of Big Data* 11, 1 (10 Jul 2024), 93. <https://doi.org/10.1186/s40537-024-00953-2>
- [7] Sharon Hirsch, Slava Novgorodov, Ido Guy, and Alexander Nus. 2021. Generating Tips from Product Reviews. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining*. Association for Computing Machinery, New York, NY, USA, 310–318. <https://doi.org/10.1145/3437963.3441755>
- [8] Mingqing Hu and Bing Liu. 2004. Mining and summarizing customer reviews. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Association for Computing Machinery, New York, NY, USA, 168–177. <https://doi.org/10.1145/1014052.1014073>
- [9] Shih-Wen Huang, Pei-Fen Tu, Wai-Tat Fu, and Mohammad Amanzadeh. 2013. Leveraging the crowd to improve feature-sentiment analysis of user reviews. In *Proceedings of the 2013 International Conference on Intelligent User Interfaces (Santa Monica, California, USA) (IUI '13)*. Association for Computing Machinery, New York, NY, USA, 3–14. <https://doi.org/10.1145/2449396.2449400>
- [10] Yeong Un Lee, Seung Hyun Chung, and Joon Young Park. 2024. Online Review Analysis from a Customer Behavior Observation Perspective for Product Development. *Sustainability* 16, 9 (2024). <https://doi.org/10.3390/su16093550>
- [11] Aytug Onan. 2021. Sentiment analysis on product reviews based on weighted word embeddings and deep neural networks. *Concurrency and Computation: Practice and Experience* 33, 23 (2021), e5909. <https://doi.org/10.1002/cpe.5909> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/cpe.5909> e5909 CPE-20-0130.R1.
- [12] Ashwin Perti, Amit Sinha, and Ankit Vidyarthi. 2024. Cognitive Hybrid Deep Learning-based Multi-modal Sentiment Analysis for Online Product Reviews. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 23, 8, Article 113 (aug 2024), 14 pages. <https://doi.org/10.1145/3615356>
- [13] Jiayin Qi, Zhenping Zhang, Seongmin Jeon, and Yanquan Zhou. 2016. Mining customer requirements from online reviews: A product improvement perspective.

Information Management 53, 8 (2016), 951–963. <https://doi.org/10.1016/j.im.2016.06.002> Big Data Commerce.

[14] Lili Shi, Jun Lin, and Guoquan Liu. 2023. Product feature extraction from Chinese online reviews: application to product improvement. *RAIRO-Oper. Res.* 57, 3 (2023), 1125–1147. <https://doi.org/10.1051/ro/2023046>

[15] Viju Sudhi, Sinchana Ramakanth Bhat, Max Rudat, and Roman Teucher. 2024. RAG-Ex: A Generic Framework for Explaining Retrieval Augmented Generation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Washington DC, USA) (SIGIR '24). Association for Computing Machinery, New York, NY, USA, 2776–2780. <https://doi.org/10.1145/3626772.3657660>

[16] Qing Sun, Jianwei Niu, Zhong Yao, and Hao Yan. 2019. Exploring eWOM in online customer reviews: Sentiment analysis at a fine-grained level. *Eng. Appl. Artif. Intell.* 81, C (may 2019), 68–78. <https://doi.org/10.1016/j.engappai.2019.02.004>

[17] Anning Wang, Qiang Zhang, Shuangyao Zhao, Xiaonong Lu, and Zhanglin Peng. 2020. A review-driven customer preference measurement model for product improvement: sentiment-based importance–performance analysis. *Information Systems and e-Business Management* 18, 1 (01 Mar 2020), 61–88. <https://doi.org/10.1007/s10257-020-00463-7>

[18] Zhentao Xu, Mark Jerome Cruz, Matthew Guevara, Tie Wang, Manasi Deshpande, Xiaofeng Wang, and Zheng Li. 2024. Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Washington DC, USA) (SIGIR '24). Association for Computing Machinery, New York, NY, USA, 2905–2909. <https://doi.org/10.1145/3626772.3661370>

[19] Cheng Yang, Lingang Wu, Kun Tan, Chunyang Yu, Yuliang Zhou, Ye Tao, and Yu Song. 2021. Online User Review Analysis for Product Evaluation and Improvement. *Journal of Theoretical and Applied Electronic Commerce Research* 16, 5 (2021), 1598–1611. <https://doi.org/10.3390/jtaer16050090>

[20] Chenxi Zhang and Zeshui Xu. 2024. Gaining insights for service improvement through unstructured text from online reviews. *Journal of Retailing and Consumer Services* 80 (2024), 103898. <https://doi.org/10.1016/j.jretconser.2024.103898>